Long-Only Equity Capacity Methodology

Distributional Liquidity Migration, Execution Horizons and Alpha Decay

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Abstract

This document defines two complementary implementations of the methodology for estimating the capacity of a long-only equity strategy. It separates the methodology from portfolio-specific runs. Each run should produce a separate report containing the supplied inputs, fitted parameters, diagnostics, scenarios, thresholds and warnings. Appendices A–C define the user input contract and distinguish supplied, policy and fitted parameters.

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*with the help of ChatGTP

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1 Framework overview

The framework maps assets under management (AUM) into a distribution of parent-order participation, execution horizons, implementation costs and captured alpha:

$$A \to F_A(p) \to D(p) \to \{\bar{c}(A), R_g(A)\} \to R_{\text{net}}(A), IR_{\text{net}}(A). \tag{1.1}$$

Turnover T is one-way annual turnover, so purchases and sales are approximated by $2TA$. In the ADV branch this is intended trading before an execution-horizon constraint. Observed cost is expressed in basis points per executed notional; the ADV scenario cost below includes completion and is expressed per intended traded dollar.

The model consumes either aggregate ADV buckets or, preferably, parent-order data. In the Burr implementation, aggregate bucket shares are shares of executed notional. The ADV-calibrated implementation jointly fits parent-order counts and traded values, using universe ADV percentiles as a liquidity prior. These are distinct weighting measures and are reported separately.

2 Strategy economics

At current AUM A_0, the key quantities are gross relative return R_g, tracking error TE, turnover T and average holdings N:

$$IR_g = \frac{R_g}{TE}, \qquad \alpha_{\text{holding}} = \frac{R_g}{N}, \qquad \alpha_{\text{cycle}} = \frac{R_g}{T}, \qquad \mathcal{H} \approx \frac{1}{T}. \tag{2.1}$$

Here $\mathcal{H}$ denotes the approximate holding period in years; H below denotes the number of holdings. Current gross alpha is treated as already reflecting the current implementation schedule. Alpha decay at alternative AUM levels is therefore measured relative to the current schedule.

3 Distributional liquidity migration

Capacity depends on parent-order participation $p = Q/\text{ADV}$, not ADV in isolation. Modelling ADV alone would require a joint model for parent-order size and its dependence on ADV.

3.1 Burr reference model

The aggregate implementation fits a Burr XII distribution:

$$F(p; c, d, \lambda) = 1 - \left[1 + \left(\frac{p}{\lambda}\right)^c\right]^{-d}. \tag{3.1}$$

It offers a flexible body and Pareto-like survival tail $1 - F(p) \sim p^{-cd}$. Empirical work finds power-law tails in trade-size distributions [1]. The calibration constrains $cd \geq 1.25$, preserving a heavy tail while ensuring finite mean participation.

The distribution changes with AUM through

$$\lambda(A) = \lambda_0 \left(\frac{A}{A_0}\right)^{\eta}, \qquad d(A) = d_0 \left(\frac{A}{A_0}\right)^{-\kappa}. \tag{3.2}$$

The scale elasticity η captures how participation grows with AUM. Tail drift κ is reserved for structural changes in liquidity composition. Bucket shares follow smoothly from

$$w_i(A) = F_A(u_i) - F_A(\ell_i). \tag{3.3}$$

The participation distribution enters the capacity calculation through its quantile representation. The distribution is weighted by executed notional: each equal-width quantile interval represents

an equal fraction of traded value, not an equal number of trades. At each AUM, set $F_A(p) = F(p; c, d(A), \lambda(A))$. Its inverse turns a fixed rank $q \in (0,1)$ into participation:

$$p_A(q) = F_A^{-1}(q) = \lambda(A)\left[(1-q)^{-1/d(A)} - 1\right]^{1/c}. \tag{3.4}$$

The same rank is used at current and alternative AUM. This is a rank-preserving scenario assumption, not an observation that the same individual orders recur. It supplies the paired participations needed for the horizon ratio in equation (6.2). When $\kappa = 0$, $p_A(q) = p_{A_0}(q)(A/A_0)^{\eta}$; with tail drift, the full inverse in equation (3.4) is required.

Bucket weights in equation (3.3) summarize this distribution for charts. The continuous model does not multiply them into its costs a second time: the notional weights are already represented by the quantile average made explicit in Section 8.

3.2 Joint ADV and parent-order calibration

Let n_b and V_b be the annual parent-order count and traded dollars in participation bucket $b = (\ell_b, u_b]$. Let $A_\star$, $H_\star$ and $T_\star$ denote the fixed historical AUM, average holdings and one-way turnover. These historical quantities are distinct from editable scenario inputs. The sample determines

$$N_\star = \sum_b n_b, \qquad V_\star = \sum_b V_b, \qquad T_\star = \frac{V_\star}{2A_\star}, \qquad \bar{Q}_b = \frac{V_b}{n_b}. \tag{3.5}$$

Universe ADV quantiles describe the cross-section of eligible stocks, not the frequency or value of trades. For supplied percentile fractions r_j and dollar ADVs L_j, interpolate between successive points in log dollars:

$$L(r) = \exp\{(1-t)\log L_j + t \log L_{j+1}\}, \qquad t = \frac{r - r_j}{r_{j+1} - r_j}. \tag{3.6}$$

Let U be the number of eligible stocks. The Streamlit implementation converts the continuous percentile curve into a finite stock-count grid with

$$M = \min(U, 1000), \qquad r_m = \frac{m - 1/2}{M}, \quad m = 1, \ldots, M. \tag{3.7}$$

Thus universes of at most 1,000 names use one midpoint rank per eligible stock; larger universes use a 1,000-point approximation. Each rank is combined with 16 evenly spaced interior participation points within each bucket. A candidate cell k in bucket b has ADV L_k, participation $p_k \in (\ell_b, u_b)$ and ticket $Q_k = p_k L_k$. Its positive prior weight is proportional to

$$\pi_{bk} \propto \exp\left\{\beta(2r_k - 1) - \frac{1}{2}\left[\frac{\log(Q_k/\bar{Q}_b)}{1.5}\right]^2\right\}. \tag{3.8}$$

The fixed log-ticket width 1.5 is a modelling choice. The liquidity-preference parameter $\beta \in \{-2, 0, 2\}$ favours illiquid, neutral or liquid ranks before calibration. For each bucket, minimum-relative-entropy weights satisfy

$$\min_{w_{bk} \geq 0} \sum_k w_{bk} \log \frac{w_{bk}}{\pi_{bk}}, \qquad \sum_k w_{bk} = 1, \qquad \sum_k w_{bk} Q_k = \bar{Q}_b. \tag{3.9}$$

The implementation solves the one-dimensional exponential tilt

$$w_{bk}(\theta_b) = \frac{\pi_{bk} \exp(\theta_b Q_k / Q_{b,\max})}{\sum_h \pi_{bh} \exp(\theta_b Q_h / Q_{b,\max})}, \qquad \nu_{bk} = n_b w_{bk}. \tag{3.10}$$

Thus $\sum_k \nu_{bk} = n_b$ and $\sum_k \nu_{bk} Q_k = V_b$. Fractional counts are quadrature weights, not claims that fractional historical orders occurred. A bucket whose mean ticket lies outside the candidate support is reported as infeasible. Changing preference refits the weights. The observations do not uniquely identify the joint distribution: the prior resolves an underidentified allocation. It does not impose equal trading shares across universe deciles or reconstruct the actual 40-stock portfolio.

3.3 Scenario scaling and migration

For scenario AUM A, holdings H and decimal one-way turnover T, the ADV model requires $1 \le H \le U$ and uses

$$\begin{aligned} r_A &= \frac{A}{A_\star}, \qquad v(A) = v_0 r_A^{-\lambda_{\text{liq}}}, \\ s_Q &= \frac{A}{A_\star}\frac{H_\star}{H}, \qquad s_N = \frac{H}{H_\star}\frac{T}{T_\star}, \\ Q'_k &= s_Q Q_k, \qquad \nu'_k = s_N \nu_k, \qquad p'_k = \frac{Q'_k}{v(A)L_k}. \end{aligned} \tag{3.11}$$

Here $v_0 > 0$ is the market-volume factor at the reference AUM (1 for observed ADV, 0.5 for a 50% volume stress). The optional liquidity-deterioration elasticity $\lambda_{\text{liq}} \ge 0$ is a user-approved stress assumption. Setting it to zero preserves the fixed-liquidity baseline; positive values reduce effective ADV as AUM grows and imply $p'_k \propto r_A^{1+\lambda_{\text{liq}}}$ when holdings are fixed. A single historical AUM sample cannot estimate this elasticity. Higher turnover increases frequency, rather than ticket size, under this convention. Higher holdings decreases tickets and increases frequency. These behavioural assumptions imply

$$\sum_k \nu'_k Q'_k = s_N s_Q V_\star = 2AT. \tag{3.12}$$

For positive turnover, count and intended-notional migration shares are

$$f_b(A) = \frac{\sum_k \nu'_k \mathbf{1}_{\{p'_k \in b\}}}{\sum_k \nu'_k}, \qquad w_b(A) = \frac{\sum_k \nu'_k Q'_k \mathbf{1}_{\{p'_k \in b\}}}{\sum_k \nu'_k Q'_k}. \tag{3.13}$$

At zero turnover the interface displays zero trading shares. It continues to show hypothetical order-level execution diagnostics. An additional bucket retains participation above 100% ADV. The ADV engine does not apply Burr elasticities η or κ; one annual sample cannot identify their response to AUM growth. Frequency changes alone leave percentage migration unchanged.

4 Impact model

Expected and realised costs are fitted separately using

$$c(p) = a + bp^\gamma, \tag{4.1}$$

subject to positive parameter bounds. Square-root impact is an important benchmark, while the implementation permits the exponent to be fitted [2]. At current AUM, the continuous distributional expectation is calibrated back to the observed notional-weighted bucket cost.

5 Empirical backtest IR curve

Optimised historical backtests at several assumed asset levels provide a second, independent view of capacity. For observations (A_i, IR_i), the dashboard fits

$$IR_{\text{BT}}(A) = a - bA^\gamma, \qquad b \ge 0, \quad 0.2 \le \gamma \le 3. \tag{5.1}$$

For each candidate γ on a fine grid, a and b are estimated by ordinary least squares; the candidate with the smallest squared residual is retained. The interface reports R^2, the fitted IR at current AUM, the zero-IR crossing $(a/b)^{1/\gamma}$ and the AUM at the selected minimum IR, $[(a - IR_{\min})/b]^{1/\gamma}$, when those crossings exist.

The Swiss example uses IR observations of 0.60, 0.53, 0.47, 0.40, 0.33, 0.27, 0.20, 0.14, 0.07 and 0.00 at AUM levels from USD 0 billion through USD 9 billion. These points give approximately

$$IR_{\text{BT}}(A) = 0.5991 - 0.06624A^{1.000}, \qquad R^2 = 0.9998. \tag{5.2}$$

This empirical curve is displayed alongside the capacity analysis but is not inserted into the execution-cost equations. Combining the two mechanically would double count capacity effects that may already be embedded in the optimised backtests. Differences between the empirical and execution-based curves are therefore treated as model-validation evidence.

6 Execution horizon

In the Burr reference model, for target daily participation ρ and maximum horizon $D_{\max}$,

$$\tau(p) = \min\left(D_{\max}, \max\left(1, \frac{p}{\rho}\right)\right). \tag{6.1}$$

The continuous duration τ drives impact, alpha capture and capacity curves. The integer $D^{\text{req}}(p) = \max(1, \lceil p/\rho \rceil)$ is retained for execution-bucket labels and threshold diagnostics. Current costs already embed the current schedule, so counterfactual effective participation is

$$p_{\text{eff},j}(A) = p_j(A)\frac{\tau_j(A_0)}{\tau_j(A)}. \tag{6.2}$$

The rule exposes the cost-versus-delay trade-off studied in optimal execution [3]. A production version can replace it with direct optimisation.

In the ADV-calibrated model, the live parent-order sample is treated as observed one-day execution. Let $p_{k,\star}$ denote order k's calibrated participation at current AUM and define its demonstrated daily capacity as

$$\rho_k^\star = \max(\rho, p_{k,\star}). \tag{6.3}$$

Thus an order historically completed at 50–100% ADV is not retrospectively divided into days by a 20% assumption. The selected ρ is a minimum capacity for orders whose demonstrated rate is lower. Required days, scheduled days and completion fraction are

$$D_k^{\text{req}} = \max(1, \lceil p_k'/\rho_k^\star \rceil), \qquad \tau_k = \min(D_{\max}, \max(1, p_k'/\rho_k^\star)), \qquad e_k = \min(1, D_{\max}\rho_k^\star/p_k'). \tag{6.4}$$

The implementation floors $D_{\max}$ to a positive integer. The dashboard's average-execution statistic is the count-weighted fractional completion time

$$\bar{\tau}(A) = \frac{\sum_k \nu_k'(p_k'/\rho_k^\star)}{\sum_k \nu_k'}. \tag{6.5}$$

It may therefore be below one day when most parent orders use less than one daily participation allowance. The integer D_k^{req} is used for multi-day classifications, while continuous τ_k drives alpha decay and removes artificial jumps at integer-day boundaries. Impact uses $\widetilde{p}_k = \min(p_k', \rho_k^\star)$ as a daily-rate proxy. This proxy approximates a participation-capped schedule and does not model the smaller final slice separately. Alpha capture uses the analytic continuous-duration extension at τ_k for the completed fraction. These two scheduling approximations should be refined together when daily executions are available.

7 Alpha decay

Let H_α be signal half-life in trading days and $\delta = \log(2)/H_\alpha$. For equal daily slices over integer D days, average alpha capture is

$$g(D, H_\alpha) = \frac{1}{D}\sum_{t=0}^{D-1} e^{-\delta t} = \frac{1 - e^{-\delta D}}{D(1 - e^{-\delta})}. \tag{7.1}$$

The dashboard evaluates the same closed form at real-valued $D = \tau$ to produce continuous scenario curves; it agrees with the discrete expression at every integer duration. For the Burr model, normalising each notional rank to current implementation gives

$$R_g(A) = R_g(A_0)\,\mathbb{E}\left[\frac{g(\tau_j(A), H_{\alpha,j})}{g(\tau_j(A_0), H_{\alpha,j})}\right]. \tag{7.2}$$

Aggregate runs currently weight alpha by executed notional. Trade-level runs should weight by each order's forecast alpha contribution. Predictors with different mean-reversion speeds imply different desired trading speeds [4].

For the ADV model, the completed alpha fraction is $e_k g(\tau_k, H_\alpha)$; unfinished notional contributes zero captured alpha. The engine normalises the aggregate capture to its historical aggregate, rather than averaging the individual ratios used by Burr. The resulting equation is specified below.

Interpretive note: where decay is applied. Alpha decay is evaluated parent order by parent order, not applied directly to portfolio IR. A one-day order has $g(1, H_\alpha) = 1$ and therefore no delay loss under this convention. Orders with $\tau > 1$ have $g(\tau, H_\alpha) < 1$, while the ADV branch also removes the alpha associated with unfinished notional through e_k. These order-level capture terms are aggregated into after-delay gross alpha. Market-impact drag is then deducted to obtain net alpha, and net alpha is divided by tracking error to obtain net IR. Because capture is normalised to the historical current-AUM implementation, the capacity curve measures additional delay loss relative to that reference rather than the absolute delay already embedded in the historical portfolio.

8 Net capacity

8.1 Burr implementation

The participation distribution specified in Section 3 determines $p_A(q)$ through equations (3.1), (3.2) and (3.4). For each notional rank q, the execution rule in equation (6.1) is evaluated at current and alternative AUM:

$$\begin{aligned} D_A(q) &= D(p_A(q))\,, \qquad D_0(q) = D(p_{A_0}(q))\,, \\ p_{\text{eff},A}(q) &= p_A(q)\frac{D_0(q)}{D_A(q)}. \end{aligned} \tag{8.1}$$

Thus distributional migration has two channels: it changes effective participation and therefore impact, and it changes execution days and therefore alpha capture.

Let $\bar{c}_{\text{obs},0} = \sum_i w_{i,0}^{\text{obs}} c_i^{\text{obs}}$ be the observed current notional-weighted cost in basis points. The impact function $c(p)$ in equation (4.1) is the raw fitted curve. Its baseline calibration multiplier is

$$K = \frac{\bar{c}_{\text{obs},0}}{\displaystyle\int_0^1 c(p_{A_0}(q))\; dq}. \tag{8.2}$$

Expected and realised curves each use their own multiplier. Making the expectation explicit, the expected cost and captured gross alpha are

$$\bar{c}(A) = K \int_0^1 c(p_{\text{eff},A}(q))\; dq, \tag{8.3}$$

$$R_g(A) = R_g(A_0) \int_0^1 \frac{g(D_A(q), H_\alpha)}{g(D_0(q), H_\alpha)}\, dq. \tag{8.4}$$

Here g is equation (7.1), and a common half-life is used for the aggregate model. These integrals are the notional-weighted expectations previously written as $\mathbb{E}[\cdot]$. Integrating over uniform ranks is

equivalent to integrating over the participation distribution supplied by Section 3, with current and alternative horizons paired at the same rank.

The expectations are evaluated numerically using J equally weighted midpoint quantiles:

$$\begin{aligned} q_j &= \frac{j-\frac{1}{2}}{J}, \quad j=1,\dots,J, \qquad K_J = \frac{\bar{c}_{\text{obs},0}}{J^{-1}\sum_{j=1}^{J} c(p_{A_0}(q_j))}, \\ \bar{c}_J(A) &= \frac{K_J}{J}\sum_{j=1}^{J} c(p_{\text{eff},A}(q_j)), \\ R_{g,J}(A) &= \frac{R_g(A_0)}{J}\sum_{j=1}^{J}\frac{g(D_A(q_j),H_\alpha)}{g(D_0(q_j),H_\alpha)}. \end{aligned} \tag{8.5}$$

The factor $1/J$ supplies the notional weight. There is no additional bucket-share multiplier. Using the same numerical grid in the calibration denominator and cost average ensures $\bar{c}_J(A_0) = \bar{c}_{\text{obs},0}$ and $R_{g,J}(A_0) = R_g(A_0)$. The continuous impact expectation requires $c\,d(A) > \gamma$ for the Burr tail; the finite-mean participation constraint alone does not guarantee this for every fitted exponent or AUM. Finite-grid runs should check sensitivity to J, particularly under tail thickening.

The resulting impact estimate is converted from cost per executed notional into annual cost per unit of portfolio capital using one-way turnover:

$$C(A) = \frac{2T\bar{c}(A)}{10{,}000}. \tag{8.6}$$

The conversion by 10,000 changes basis points into a decimal return. Net alpha and IR are

$$R_{\text{net}}(A) = R_g(A) - C(A), \qquad IR_{\text{net}}(A) = \frac{R_{\text{net}}(A)}{TE}. \tag{8.7}$$

The quantities R_g, C and TE are expressed as decimal annual returns, and T as decimal one-way turnover. Distributional migration therefore affects both components of net return: impact in $C(A)$ and delay in $R_g(A)$. Tracking error and turnover remain fixed assumptions in this specification.

For example, consider a notional rank whose participation increases from 16% to 32% ADV. With a 20% minimum daily capacity and a non-binding horizon cap, its horizon increases from one to two days. Effective participation becomes $32\% \times 1/2 = 16\%$, so its modeled impact is unchanged, while its alpha-capture ratio becomes $g(2,H_\alpha)/g(1,H_\alpha) < 1$. Migration can therefore lower net IR through delay even when scheduling offsets the impact increase. This is an illustrative rank, not a claim about the entire portfolio.

A run report should show the complete capacity curve and explicitly selected thresholds rather than presenting a single universal capacity number.

8.2 ADV-calibrated implementation

Section 3.2 supplies calibrated synthetic counts and tickets. Section 3.3 scales them into participation, and equation (6.4) supplies completion and delay. Because every ticket scales by the same factor, the intended-notional weights are invariant:

$$\omega_k = \frac{\nu_k Q_k}{V_\star}, \qquad \sum_k \omega_k = 1. \tag{8.8}$$

Evaluate the historical reference at $A = A_\star$, $H = H_\star$, $v = 1$, under the demonstrated one-day capacities, horizon and half-life. Use a subscript $\star$ for those reference execution quantities. Then

$$G_\star = \sum_k \omega_k e_{k,\star} g(\tau_{k,\star}, H_\alpha), \qquad K_{\text{ADV}} = \frac{\sum_b (V_b/V_\star) c_b^{\text{obs}}}{\sum_k \omega_k e_{k,\star} c(\widetilde{p}_{k,\star})}. \tag{8.9}$$

The current dashboard uses expected bucket costs for this anchor. Its captured gross alpha and cost in basis points per intended traded dollar are

$$\begin{aligned} R_g^{\mathrm{ADV}}(A) &= R_{g,\star}\frac{\sum_k \omega_k e_k g(\tau_k, H_\alpha)}{G_\star}, \\ \bar{c}^{\mathrm{ADV}}(A) &= K_{\mathrm{ADV}} \sum_k \omega_k e_k c(\widetilde{p}_k), \\ U(A) &= \sum_k \omega_k (1 - e_k). \end{aligned} \tag{8.10}$$

Here U is the unfinished fraction of intended notional. Substituting $\bar{c}^{\mathrm{ADV}}$ into equations (8.6) and (8.7) yields annual drag, net alpha and net IR. In this branch $2AT$ is intended annual trading; completion is already included through e_k, and must not be applied again. Declining costs caused by non-completion are not evidence of improved execution: U and captured alpha must be read alongside costs.

Interpretive note: after-delay versus net alpha. After-delay alpha contains the loss from slower implementation but has not yet deducted market impact. Net alpha subtracts annual trading-cost drag. In the default Swiss reference case, $R_{g,\star} = 0.5000\%$, captured alpha is 100% by reference normalisation, expected impact is 8.69 bp and $T = 13.9\%$. Equation (8.6) gives 2.42 bp of annual drag, so after-delay alpha is 0.5000% and net alpha is approximately 0.4758%. Positive implementation cost therefore places the net-alpha curve below the after-delay curve even at the reference AUM.

At the historical reference the normalisation reproduces the supplied alpha and weighted cost. Changing execution-policy controls also changes $G_\star$ and K_{ADV}. Consequently the current engine compares relative AUM scenarios under each selected policy; it is not an absolute historical execution-policy backtest. Neither aggregate branch identifies overlapping orders in one stock, time-varying volume, actual holdings selection or order-level alpha contributions.

9 Required run-report contents

Every portfolio run report must include:

1. run identity, timestamp, code version and input source;
2. supplied strategy, liquidity, execution and alpha-decay inputs;
3. fitted distribution and impact parameters with fit diagnostics;
4. current-AUM reconciliation to observed costs and alpha;
5. AUM scenario table and continuous capacity curves;
6. engine identity, count and dollar calibration residuals, migration and execution-horizon diagnostics;
7. for ADV runs, preference prior, volume stress, uncapped required days and unfinished notional;
8. selected capacity thresholds and exact crossing estimates;
9. data-quality, extrapolation and model-risk warnings.

10 Validation requirements

A valid run should confirm that notional shares sum to one, AUM and half-life are positive, participation buckets are ordered and non-overlapping, expected and realised costs share a common benchmark, and current-AUM impact reconciles to the supplied weighted cost. The report should flag missing trade-level alpha weights, large extrapolations, horizons at $D_{\max}$, unstable tail fits and sensitivity to user policy parameters.

11 Current calibration and dashboard interpretation

The supplied annual sample contains 971 parent orders and USD 944.475 million of two-way traded value at historical AUM USD 3.4 billion. It implies $T_\star = 13.889338\%$ one-way turnover. The dashboard's rounded 13.9% scenario produces USD 945.2 million of intended annual trading and approximately 971.7 expected orders at 40 holdings. This rounding difference is intentional; the calibration table itself reconciles to the exact historical totals.

The 203-stock eligible universe has a one-year ADV lookback. It therefore uses 203 midpoint stock ranks in equation (3.7). The supplied percentile boundaries are shown below; they are not mean ADVs within deciles.

Q0	Q10	Q20	Q30	Q40	Q50	Q60	Q70	Q80	Q90	Q100
0.020	0.050	0.130	0.800	1.000	1.420	2.000	4.000	6.000	50.000	200.000

ADV in USD millions per day.

At a 20% minimum daily capacity, equation (6.5) gives approximately 0.08 trading day for the calibrated current-AUM parent orders and 0.16 day at twice current AUM. These are count-weighted fractional completion times. Every live current-AUM parent order is classified as a one-day execution; whole-day buckets begin to migrate only as orders scale beyond their demonstrated capacities.

Worked migration example. Hold the Swiss strategy at 40 holdings, 13.9% one-way turnover and observed market ADV. Increasing AUM from USD 3.4 billion to USD 6.8 billion gives $s_Q = 2$ and $s_N = 1$ in equation (3.11). Every synthetic parent order doubles in size, the expected annual parent-order count remains approximately 972, and intended annual trading doubles from about USD 945.2 million to USD 1.8904 billion. Because ADV is unchanged, each order's participation also doubles: an order at 0.7% ADV moves to 1.4%, an order at 3% moves to 6%, and an order at 40% moves to 80%. Reassigning every calibrated order produces the following migration.

	Parent-order count share		Traded-dollar share	
Participation bucket	USD 3.4bn	USD 6.8bn	USD 3.4bn	USD 6.8bn
0–1% ADV	82.49%	29.47%	28.91%	6.72%
1–5% ADV	11.64%	56.36%	26.01%	27.09%
5–10% ADV	3.40%	8.30%	17.98%	21.10%
10–25% ADV	1.85%	3.70%	16.58%	20.14%
25–50% ADV	0.51%	1.56%	9.18%	14.41%
50–100% ADV	0.10%	0.51%	1.34%	9.18%
>100% ADV	0.00%	0.10%	0.00%	1.34%

The number of orders does not increase in this example; the same frequency is redistributed toward higher participation buckets. Count shares and dollar shares differ because a small number of large orders can represent a disproportionate share of traded value. Changing holdings or turnover would activate the additional scaling channel s_N rather than this pure AUM-size experiment.

With 40 assumed holdings and unchanged frequency, doubling AUM doubles every synthetic ticket; annual order count remains unchanged. With the same AUM and twice the holdings, tickets

halve and frequency doubles. With twice the turnover alone, frequency doubles but participation shares are unchanged. These identities are model assumptions, not estimated portfolio responses.

The main dashboard provides Burr as a comparison to the strategy's saved migration default. The liquidity demo provides independent ADV and Burr controls initialized from the selected strategy. The ADV demo shows count and dollar migration, historical reconciliation, execution feasibility and relative alpha capture. The impact demo illustrates $a + bp^{\gamma}$ independently. The alpha-decay demo adds a waiting-time factor $2^{-t/H_{\alpha}}$ to implemented gross alpha and displays 126 trading days, approximately six months; this is an explanatory extension, not an additional waiting penalty in the capacity engine. The Strategy creator edits and persists strategy-specific AUM–IR observations together with all other inputs. The Backtest IR curve tab initializes a, b and γ from equation (5.1); the Capacity Lab sidebar can adjust them for the active analysis. An independent backtest demo exposes the same three parameters and starts from the selected strategy's fit. The empirical curve remains separate from modeled net IR. The Summary area places the two capacity indications side by side, reporting implied IR at current AUM, AUM at the selected minimum IR and zero-IR AUM. Execution-model crossings are evaluated within the selected chart horizon, while empirical backtest crossings are solved directly from a, b and γ.

Validation covers historical counts and values, percentile interpolation, annual trading conservation, volume-stress equivalence and infeasible calibration inputs. Quadrature density, liquidity preference and ticket-prior width require sensitivity review before operational use.

12 Running the dashboard implementations

The methodology is distributed through two live dashboard implementations. The TypeScript browser version and the Python Streamlit version both evaluate the model interactively from the source parameters. Both implementations use USD, the same live bucket observations, the same universe ADV quantiles and the same Burr and ADV calibration inputs.

12.1 Browser dashboard

The browser dashboard requires Node.js 22.13 or later. From the project root, install the locked dependencies and start the local development server with

```
cd dashboard
npm ci
npm run dev -- --host 127.0.0.1 --port 3011
```

The dashboard is then available at `http://localhost:3011`. The process remains attached to the terminal and is stopped with `Ctrl+C`. Before distributing a change, the production build and the model checks should be run from the `dashboard` directory:

```
npm run build
node --experimental-strip-types --test tests/adv-migration.test.mjs
```

The browser version recalculates immediately when the user changes AUM, holdings, turnover, participation, execution horizon, alpha half-life or the available model-specific controls.

12.2 Streamlit dashboard

The Streamlit implementation requires Python 3.11 or later. From the project root, create an isolated environment, install the declared dependencies and start the application with

```
cd streamlit
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
streamlit run app.py
```

On Windows PowerShell, the environment is activated with `.venv\Scripts\Activate.ps1`. Streamlit normally opens `http://localhost:8501`; the server is stopped with `Ctrl+C`. The Streamlit version contains the Overview, Summary, Backtest IR curve, Liquidity migration, Demo building blocks and Strategy creator areas. The built-in Swiss Strategy has 203 eligible stocks. The Strategy creator saves a named case containing dashboard defaults, liquidity buckets, universe size, ADV percentiles and backtest IR observations in the local `data/strategies.user.json` file; saving an existing name replaces its local record. The sidebar selector loads the selected case throughout the dashboard and supplies every control's default. The saved universe size constrains holdings and sets the finite grid in equation (3.7). The application reads the live bucket and ADV JSON files used by the TypeScript dashboard and recalculates scenarios when AUM, turnover, holdings, participation, horizon, half-life or model controls change. The Streamlit model checks are run from its directory with

```
python -m unittest discover -p 'test_*.py'
```

A User-supplied data inputs

Table 1: Required and optional user data

Input	Requirement	Unit / format	Definition and validation
Current AUM A_0	Required	Currency bn	Average deployed AUM represented by the liquidity sample; must be positive.
Gross expected alpha R_g	Required	Decimal p.a.	Expected relative return before modeled impact; document whether existing live slippage is embedded.
Tracking error TE	Required	Decimal p.a.	Annualised ex-ante or realised relative volatility; must be positive and methodologically aligned with alpha.
One-way turnover T	Required	Decimal p.a.	Purchases or sales divided by average AUM. Do not supply two-way turnover without changing the convention.
Average holdings N or H	Required for ADV scaling	Count	Historical and scenario average positions; distinct from eligible universe size.
Bucket lower and upper bounds	Required for aggregate run	Fraction of ADV	Ordered, non-overlapping participation intervals covering the full positive range.
Executed-notional share	Required for aggregate run	Fraction summing to 1	Annual executed-notional mass in each bucket; not trade-count frequency.
Representative participation	Required for impact fit	Fraction of ADV	Prefer the notional-weighted mean participation within each bucket.
Expected impact	Required	bp / executed notional	Expected implementation cost using a consistent arrival-price or other declared benchmark.
Realised impact	Recommended	bp / executed notional	Realised cost using the same benchmark as expected impact.
Parent-order notional or shares	Required for trade-level run	Currency / shares	Parent order before child-order slicing, net of cancellations according to a declared convention.
Security ADV	Required for trade-level run	Currency or shares/day	ADV measured consistently with order units, including declared lookback and lag.
Annual bucket counts and values	Required for ADV calibration	Count / currency	Parent orders and two-way traded dollars for the same historical period and strategy.

Input	Requirement	Unit / format	Definition and validation
Universe ADV quantiles	Required for ADV calibration	Percentile / currency per day	Include minimum and maximum, universe size and lookback; specify the daily-volume statistic.
Order timestamps	Recommended	Datetime	Decision, release, first fill and final fill times.
Actual execution horizon	Recommended	Trading days	Number of sessions over which the parent order was worked.
Forecast alpha	Recommended	bp or return	Expected contribution at the order decision timestamp.
Signal family	Recommended	Category	Used to assign or estimate signal-specific decay.
Observed alpha path	Preferred	Return by horizon	Forward alpha after 1, 2, 3, 5, 10 or more trading days for empirical half-life estimation.
Side, spread and volatility	Recommended	Category / bp / decimal	Used for richer asymmetric impact and execution-risk models.

A.1 Clarification: contribution to notional-weighted participation

The Overview statistic labelled average weighted trade size is the intended-notional-weighted mean parent-order participation. Let w_b be bucket b's share of traded dollars and let $\bar{p}_b$ be the calibrated notional-weighted mean participation within that bucket. The bucket's contribution to the portfolio average is

$$C_b = w_b \bar{p}_b, \qquad \bar{p} = \sum_b C_b. \tag{A.1}$$

Here C_b and $\bar{p}$ are expressed in percentage points of ADV when $\bar{p}_b$ is quoted as a percentage. A bucket weight is first converted from percent to a decimal before multiplication.

For the default Swiss 0–1% ADV bucket, the traded-dollar share is 28.9064%, or 0.289064, and the calibrated within-bucket mean participation is 0.7063% ADV. Its contribution is therefore

$$C_{0\text{–}1\%} = 0.289064 \times 0.7063\% = 0.2042\% \text{ ADV} \simeq 0.20 \text{ percentage points.} \tag{A.2}$$

The 0.20 percentage-point result is not 20% of the portfolio and is not the fraction of parent orders in the bucket. The source's 28.91% and 26.01% fields are traded-dollar shares for the 0–1% and 1–5% buckets. Parent-order count shares instead come from the observed counts: $801/971 = 82.49\%$ and $113/971 = 11.64\%$. Summing equation (A.1) over all calibrated buckets gives 10.3236% ADV. The result exceeds the typical order's participation because the relatively few high-participation orders carry disproportionate traded notional. A count-weighted mean answers a different question and is approximately 1.8% ADV in the same case.

A.2 Interpretation of curve shape

The Overview chart places average theoretical impact and notional-weighted parent-order participation against AUM. The two series have different shapes because they represent different parts of the model.

The fitted impact function $c(p) = a + bp^{\gamma}$ has $\gamma = 0.386 < 1$ in the Swiss calibration and is therefore concave in participation. This shape is consistent with the supplied expected-cost buckets: cost rises materially through the middle buckets but changes only from 22.45 bp to 22.48 bp between the 25–50% and 50–100% ADV buckets. The execution rule further limits daily participation to $\widetilde{p} = \min(p, \rho)$. Larger parent orders are therefore spread over more days rather than charged as if their full participation were executed in one session. Under the current case, average theoretical impact rises from approximately 8.69 bp at USD 3.4 billion to 10.28 bp at USD 15 billion. Additional pressure appears in alpha delay and unfinished notional, which should be read alongside the impact series.

In the baseline $\lambda_{\text{liq}} = 0$, weighted participation is linear in AUM under fixed holdings, market ADV and liquidity preference. Equations (3.11) and (8.8) imply

$$p'_k(A) = p_k \frac{A}{A_\star}, \qquad \bar{p}(A) = \sum_k \omega_k p'_k(A) = \bar{p}(A_\star)\frac{A}{A_\star}. \tag{A.3}$$

The linear mean does not imply that the liquidity distribution is benign. As AUM rises, individual orders cross bucket boundaries and the shares above 10%, 25%, 50% and 100% ADV can change rapidly. Tail percentiles, threshold exceedance shares, multi-day orders and unfinished notional are therefore more informative about nonlinear liquidity pressure than the mean alone.

The optional deterioration stress makes the mean itself nonlinear without changing the order-size accounting:

$$\bar{p}(A) = \bar{p}(A_\star)\left(\frac{A}{A_\star}\right)^{1+\lambda_{\text{liq}}}. \tag{A.4}$$

For example, $\lambda_{\text{liq}} = 0.20$ changes a twofold AUM increase into a 2.30-fold participation increase and a fourfold AUM increase into a 5.28-fold participation increase. The Overview tail-pressure chart separately reports traded notional above 10% and 25% ADV and the notional-weighted 90th-percentile parent-order participation.

A convex impact curve would require $\gamma > 1$ or an additional crowding, spread, volatility or liquidity-risk term and should be supported by an appropriate recalibration sample. A nonlinear mean-participation curve requires an AUM-dependent mechanism. The implemented λ_{liq} stress is a simple representation; changing holdings, endogenous migration toward less-liquid securities or binding finite-universe constraints are richer alternatives.

B User-set policy and scenario parameters

Table 2: Parameters chosen or approved by the user

Parameter	Example	Unit	Interpretation
Participation-scale elasticity η	0.85	Elasticity	Response of the Burr scale to AUM. Set to 1 for proportional scaling; estimate from portfolio rescaling where possible.
Tail-thickening elasticity κ	0.00	Elasticity	Structural change in upper-tail liquidity composition. Keep at zero unless supported by portfolio simulations or observations.
Target daily participation ρ	20%	ADV/day	Minimum execution capacity; higher demonstrated live rates are retained.
Maximum execution horizon $D_{\max}$	10	Trading days	Burr caps the horizon; ADV caps completion and reports unfinished notional.
Alpha half-life H_α	10	Trading days	Time for forecast alpha to halve. Prefer signal-specific empirical estimates.
AUM scenario grid	1–100	Currency bn	Range over which curves and thresholds are evaluated.
Liquidity preference β	$-2, 0, 2$	Prior tilt	Illiquid, neutral or liquid preference; not an observed trading allocation.
Market-volume factor v	1.00	Multiplier	Stress scenario applied to ADV after historical calibration.
Liquidity deterioration λ_{liq}	0.00	Elasticity	Optional decline in effective ADV as AUM grows. Zero preserves the linear baseline; positive values require SME approval or multi-AUM evidence.
Net IR threshold	0.40	Ratio	Minimum acceptable net information ratio.

Parameter	Example	Unit	Interpretation
Alpha-retention threshold	80%	Percentage	Minimum fraction of current gross alpha remaining after delay and impact.
Impact-drag threshold	100	bp p.a.	Maximum acceptable annual return drag.
Alpha-capture threshold	95%	Percentage	Minimum gross alpha retained after execution delay, before impact.

C Fitted parameters and outputs

The user does not normally supply Burr parameters (c, d, λ_0) or impact parameters (a, b, γ); the model estimates them. Every run report should disclose them along with the tail index cd, bucket-fit residuals, impact-fit residuals, baseline calibration multiplier, threshold crossings and warnings. Parameters should be versioned with their input sample and estimation date. ADV runs additionally disclose bucket tilts, grid density, prior width, preference, count and value residuals, and the distinction between exact historical turnover and rounded scenario inputs.

References

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